(Please write your Enrollment Number) Enrollment No. 03801018021

MID TERM EXAMINATION
(CBCS)
B. Tech (CSE/ECE), 2nd Semester
(June, 2022)

Subject Code: BAS 102	Subject: Applied Mathematics II

Time : 1 ½ Hours Maximum Marks : 30

Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1		**(3+3+4=10)**
	(a) Solve by the method of variation of parameters $(D^2 - 2D + 1)y = x^{\frac{3}{2}}e^x$	
	(b) Evaluate $\int_0^\infty \frac{e^{-t}sin^2t}{t}\,dt$ using Laplace Tansforms.	
	(c) Find the inverse Laplace transform of $\frac{\pi e^{-s}}{s^2+\pi^2}$ in terms of unit step function.	
Q2		**(6+4=10)**
	(a) Solve the equation $(x^2D^2 + xD + 4)y = Sin(2\log x)$ using operator method.	
	(b) Apply Convolution Theorem to evaluate $L^{-1}\left\{\frac{1}{(s-2)(s+2)^2}\right\}$	
Q3		**(5+5=10)**
	(a) Solve the equation $(tD^2 + 2D + t)y = \cos t$ given that $y(0) = 1$ using Laplace Transforms.	
	(b) Solve the equation $(D^3 + 2D^2 + D)y = e^{2x} + x^2 + Sin\,x$ using operator method.	
Q4		**(6+4=10)**
	(a) Solve $y'' - 3y' + 2y = x^2 + e^x$ by the method of undetermined coefficients.	
	(b) Find the inverse Laplace Transform of $\frac{1}{s(s+a)^3}$	